

(14) Evaluate:

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x + \cos x} dx$$

OR

Evaluate:

$$I = \int_0^{\frac{3}{2}} [x \cos \pi x] dx$$

(15) Find:

$$I = \int (3x + 1) \sqrt{4 - 3x - 2x^2} dx$$

(16) Solve the differential equation:

$$y + x \frac{dy}{dx} = x - y \frac{dy}{dx}$$

(17) Form the differential equation of the family of circles in the second quadrant and touching the coordinate axes.

(18) Solve the equation for x :

$$\sin^{-1} x + \sin^{-1}(1 - x) = \cos^{-1} x$$

OR

If

$$\cos^{-1} \frac{x}{a} + \cos^{-1} \frac{y}{b} = \alpha$$

prove that

$$\frac{x^2}{a^2} - 2\frac{xy}{ab}\cos\alpha + \frac{y^2}{b^2} = \sin^2\alpha$$

- (19) A trust invested some money in two type of bonds. The first bond pays 10% interest and second bond pays 12% interest. The trust received Rs. 2800 as interest. However, if trust had interchanged money in bonds, they would have got Rs. 100 less as interest. Using matrix method, find the amount invested by the trust.
- (20) There are two types of fertilisers 'A' and 'B'. 'A' consists of 12% nitrogen and 5% phosphoric acid whereas 'B' consists of 4% nitrogen and 5% phosphoric acid.
- (21) Five bad oranges are accidentally mixed with 20 good ones. If four oranges are drawn one by one successively with replacement, then find the probability distribution of number of bad oranges drawn.
- (22) Find the position vector of the foot of the perpendicular and the perpendicular distance from the point P with position vector

$$2\hat{i} + 3\hat{j} + 4\hat{k}$$

to the plane

$$\vec{r} \cdot (2\hat{i} + \hat{j} + 3\hat{k}) - 26 = 0$$

Also find image of P in the plane.

- (23) Show that the binary operation $*$ on

$$A = R - \{-1\}$$

defined as

$$a * b = a + b + ab$$

for all $a, b \in A$ is commutative and associative on A .

- (24) Prove that the least perimeter of an isosceles triangle in which a circle of radius r can be inscribed is

$$6\sqrt{3}r$$

OR

If the sum of lengths of hypotenuse and a side of a right angled triangle is given, show that area of triangle is maximum, when the angle between them is

$$\frac{\pi}{3}$$

- (25) Prove that the curves

$$y^2 = 4x$$

and

$$x^2 = 4y$$

divide the area of square bounded by

$$x = 0, x = 4, y = 4, y = 0$$

into three equal parts.

- (26) Using properties of determinants, show that $\triangle ABC$ is isosceles if

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos^2 A + \cos A & \cos^2 B + \cos B & \cos^2 C + \cos C \end{vmatrix} = 0$$

OR

A shopkeeper has 3 varieties of pens 'A', 'B' and 'C'. Meenu purchased 1 pen of each variety for a total of Rs. 21.